

MedMind AI: Final Project Report

Group 6: Yateen Sudhir Sakhare & Jay Shah - (Elderly AI2)

Project Title: MedMind AI – An Intelligent Medication Assistant with Voice*

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Abstract

The MedMind AI project was initiated to address critical issues faced by the elderly population: medication non-adherence and the risk of adverse drug interactions. Our proposed solution is a privacy-focused, voice-activated medication assistant. Over the course of the term, our team successfully designed and developed a functional prototype. This application features a curated medication knowledge base of 36 common drugs, a voice-activated interface for adding medications, and a robust reminder system with both on-screen and email notifications. A key accomplishment was our strategic pivot from a complex interaction checker to a more user-centric system that provides clear, comprehensive medication information directly from the knowledge base. This report details our journey, including our accomplishments, challenges, and the valuable lessons learned.

1. Introduction

Medication management presents significant challenges for many individuals, particularly for the elderly. The risk of missed doses, incorrect dosages, and dangerous drug-drug or drug-food interactions can lead to serious health complications. Existing solutions often lack the accessibility required for a user base that may be less tech-savvy.

MedMind AI was developed as a solution to these problems. The application's core design principles prioritize ease of use and safety. By integrating a voice interface, we aimed to lower the barrier to entry for our target users. The core functionality is built on a reliable knowledge base that provides up-to-date and accurate information, empowering users to manage their medication regimen with greater confidence.

2. Project Management and Schedule

Our project followed an iterative development schedule, allowing us to adapt to challenges as they arose.

- **Phase 1 (Weeks 1-2):** Project planning, team role definition, and initial research into technology stacks and data sources.
- **Phase 2 (Weeks 3-6):** Core component development, including building the medication knowledge base and initial UI and voice module prototypes.
- **Phase 3 (Weeks 7-9):** UI and voice integration, focusing on a complete end-to-end user flow for reminders and information display.
- **Phase 4 (Weeks 10-11):** Final testing, UI/UX refinement, and documentation.

The most significant challenge we faced was the rejection of our request for access to the DrugBank dataset. This required an immediate and decisive pivot in our project plan. Instead of using a licensed data source, we manually curated a dataset of **36 common drugs** from reputable, publicly available sources such as the

FDA's OpenFDA, Mayo Clinic, and Drugs.com. This decision, while time-consuming, allowed us to maintain full control over the data's accuracy and relevance to our target demographic.

3. System Architecture & Design

The MedMind AI application is built on a modular architecture that separates key functions.

- **User Interface (UI):** Developed by Jay, this is the user-facing component built using [Expo Native React]. It allows users to add and delete medications and view their reminder schedule.
- **Voice Module:** This consists of two primary components. **Speech-to-Text (STT)**, implemented using [SpeechRecognition library and pyaudio], converts user voice commands into text. **Text-to-Speech (TTS)**, implemented using gTTS library, provides audio feedback to the user, reading out confirmations and important information. This was handled by Yateen.
- **Medication Knowledge Base:** Curated by Yateen, this is a central .json file that serves as the application's single source of truth for medication data. It contains detailed information on 36 drugs.
- **Core Logic:** This module, also developed by Yateen, handles the processing of user requests. It performs lookups in the knowledge base and prepares data for display.
- **Reminder System:** A custom-built system developed by Jay that schedules alerts based on user-inputted times. It triggers both on-screen notifications and sends email reminders to the user. It also speaks out the reminder for user to take medicine.

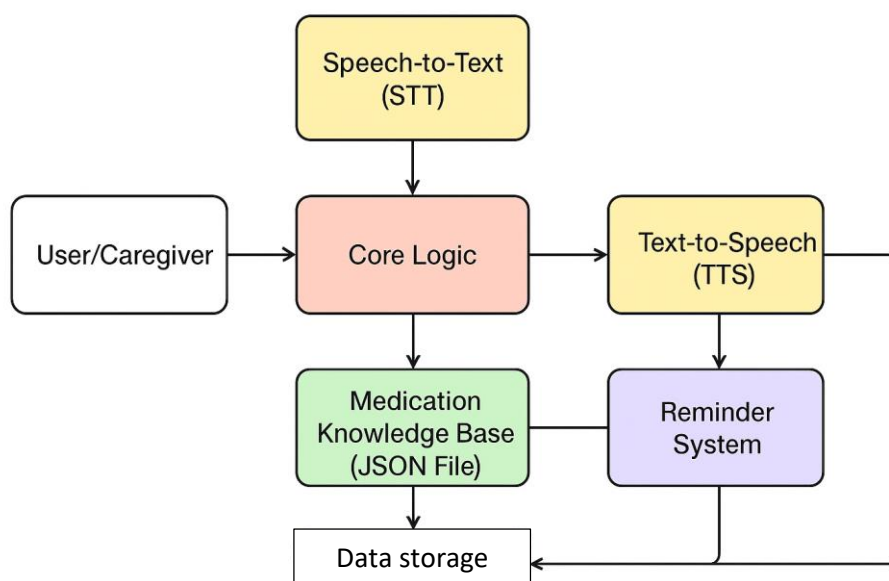


Figure 1: MedMind AI System Architecture

4. Key Features & Implementation

Our project successfully implemented several key features that form a cohesive and valuable product.

- **Curated Medication Knowledge Base:** A core accomplishment was the creation of a verified knowledge base of 36 drugs, which are commonly prescribed to elderly patients. The data includes generic and brand names, typical dosages, intended uses, and crucial interaction information.
- **Smart Information Display:** In a strategic pivot from a traditional interaction checker, our application now displays all relevant medication information directly to the user after a drug has been added. This includes potential interactions with food, alcohol, and other drugs. This design choice provides a more comprehensive and less alarmist approach, empowering the user with complete information in a single view.

- **Voice Interface:** The application's voice module enhances accessibility. The TTS feature now provides audio feedback for confirmations and reminders, making the app easier to use for individuals with visual impairments.
- **Robust Reminder System:** The app features a reliable reminder system that ensures users do not miss their medication. In addition to on-screen alerts, the system also sends a detailed email reminder to the user's registered email address.

5. Challenges and Lessons Learned

Our team encountered several challenges that became valuable learning experiences. The primary hurdle was the inability to access a licensed medication database. This led us to our most important lesson: adaptability is a critical skill in project management. By pivoting to manual data curation, we not only salvaged the project but also gained a deeper understanding of data sourcing and verification.

Another key lesson was the importance of clear communication and modular development. By working in parallel on distinct components (knowledge base vs. UI), we were able to make significant progress independently while staying synchronized through regular check-ins.

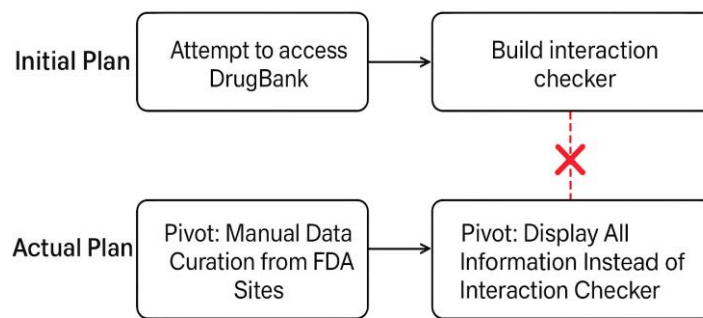


Figure 2: The Evolved Project Plan

6. Conclusion and Future Work

MedMind AI successfully delivered a functional and user-centric prototype that addresses a vital public health need. We have demonstrated a practical application of AI, voice technology, and data management in a single system. The project's success is a testament to our team's collaborative spirit and ability to overcome technical challenges.

Looking forward, the project has immense potential for expansion. Future work could include:

- Integrating with a live, publicly available pharmacy API to provide real-time pricing and availability information.
- Expanding the knowledge base to include a broader range of medications, supplements, and herbal remedies.
- Developing a caregiver portal to allow family members to monitor medication adherence remotely.

7. Reflection on the Project

Overall, we believe the project went exceptionally well. We began with a clear vision and were able to deliver a working solution despite an unexpected and significant setback. Our ability to pivot and still meet our goals is something we are particularly proud of. For future projects, we would aim to conduct more thorough research on data accessibility and licensing in the initial planning phase to avoid similar delays.

8. Project Artifacts

- **GitHub Repository Link:**
 - <https://gitlab.com/jayshah398/medmind>
 - **Contents (as of September 2, 2025) on branch Add_database_and_speech_to_text:**
 - .gitignore file (for Python virtual environment, sensitive config files).
 - README.md (initial project overview).
 - app / **directory:**
 - signup.tsx – [Sign up logic]
 - explore.tsx – [Med Database 'POST', 'DELETE' and add meds] navigation logic/Drug list modification/speech to text
 - tabs / **directory**
 - index.tsx – [Login Page logic]
 - lib
 - notification.tsx - [reminder notification logic]
 - alerts.ts -> [notification on web]
 - **Note:** These files include basic validation logic.
 - backend
 - data/ **directory:**
 - app.py - [Backend server which holds database array as well as user data including logic to convert speech to text]. This file will contain the intended Python structure for the knowledge base/server for the front end.
 - medicationkb.json
 - venv / **directory** - (Virtual environment to run the backend example python3, flask etc..)

References

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Appendix: Project Artifacts

 medicationkb.json **(Curated Knowledge Base)**

Appendix A:

 app.py **(Backend file)**

Appendix B:

 explore.tsx **(Frontend Logic file)**

Appendix C:

 signup.tsx **(Signup page logic)**

Appendix D:

 Final Presentation recording-Capstone project 1-MedmindAI.mp4 **(Demo)**

Appendix E:
